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PAIRED OBSERVATIONS:

Let's take N samples from population 1 and population 2

- Let X_{1i} show the i^{th} sample of population 1.

And

- Let X_{2i} show the i^{th} sample of population 2.

$$i = 1, 2, 3 \dots N$$

⇒ The pairing will be made as

$$X_{11} - X_{21}$$

$$X_{12} - X_{22}$$

$$\vdots$$

$$X_{1N} - X_{2N}$$

Let the difference be

- $D_i = X_{1i} - X_{2i}$

$$\bullet \bar{D} = \frac{1}{N} \sum_{i=1}^N D_i = \frac{1}{N} \sum_{i=1}^N (X_{1i} - X_{2i})$$

$$\bullet S_D^2 = \frac{1}{N-1} \sum_{i=1}^N (D_i - \bar{D})^2$$

↘
Sample mean of differences.

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Hypothesis:

$$H_0 = \mu_D = d_0$$

$$H_1 : \mu_D \neq d_0 - (1)$$

$$H_1 : \mu_D < d_0 - (2)$$

$$H_1 : \mu_D > d_0 - (3)$$

The judgement is made on true mean, variance will not be given.

So in this case t-test will be used.

$$t_0 = \frac{(\bar{D} - d_0)}{S_d / \sqrt{N}}$$

$$V = N - 1$$

Critical Region:

For case (1)

$$t \geq t_{\alpha/2} \text{ or } t \leq -t_{\alpha/2}$$

For case (2)

$$t \leq -t_{\alpha}$$

For case (3)

$$t \geq t_{\alpha}$$

Question: Paired Sample t-Test

A researcher conducted a study on 15 wild deer to examine whether the drug succinylcholine affects androgen levels in their blood. Androgen levels (in ng/mL) were measured **before** and **30 minutes after** the injection. The following summary statistics were obtained for the differences in androgen levels (After - Before):

- Sample Mean Difference ($\bar{d}$): 9.848
- Sample Standard Deviation (s_e): 18.474
- Sample Size (n): 15
- Significance Level (α): 0.05

Using a **paired sample t-test**, test whether there is a significant change in androgen levels after the injection.

Should the null hypothesis be rejected?

Data:
 $N = 15$
 X_{1i}, X_{2i}, D_i 's given
 $\bar{D} = 9.848$
 $std = 18.747$
 $\alpha = 0.05$

Sol:
Hypothesis:
 $H_0 = \mu_D = 0$
 $H_1 = \mu_D \neq 0$

Test statistics:
$$T = \frac{\bar{D} - d_0}{s_d / \sqrt{N}}$$
$$= \frac{9.848 - 0}{18.747 / \sqrt{15}}$$
$$= 2.06$$

Critical Region:
 $t > t_{\alpha/2} \text{ or } t < -t_{\alpha/2}$
 $t_{\alpha/2} = 2.145, -t_{\alpha/2} = -2.145$

Conclusion:
 H_0 Accepted as its value is outside critical region.

